

After completing the preparation materials, students should be able to:

1. Compare and contrast the pharmacokinetic properties of the anti-Herpesvirus drugs.
2. Correlate the structure-activity relationships, mechanisms of action, and mechanisms of viral resistance with specifically targeted viral molecular processes and give examples.
3. Explain the relative selectivity for virus-infected cells versus non-virus infected cells for each of the antiherpes drugs.
4. Defend the clinical rationales for the selection of specific antiviral drugs and give examples of their clinical uses.
5. Relate the pharmacological properties of the various drugs to the mechanisms of adverse effects, drug-disease interactions, and drug-drug interactions.

What you need to know and understand

- Herpes viruses are associated with a broad spectrum of diseases, such as cold sores, viral encephalitis, and genital infections.
- Antiherpes agents exert their actions during the acute phase of viral infections. They are not effective during the latent phase.
- Most of the drugs that are effective against herpes viruses are nucleoside analogs. They are antimetabolites activated by viral or host cell kinases to their triphosphorylated (-TP) forms, which inhibit viral polymerases.
- As a class, they are formulated for oral, intravenous and topical use. They have short plasma half-lives and are eliminated by renal excretion. Dosage should be reduced in patients with renal impairment.
- Acyclovir, valacyclovir (acyclovir prodrug), and famciclovir (penciclovir prodrug) are converted to the drug-monophosphate by viral thymidine kinase, then to the drug-triphosphate by host kinases. They are for the systemic treatment and prophylaxis of herpes simplex HSV-1 and HSV-2 and varicella zoster (chicken pox and shingles) infections.

What you need to know and understand

- Intravenous/high dose acyclovir toxicities include nephrotoxicity and neurotoxicity (delirium, tremors, seizures, etc). Very high dose valacyclovir is associated with thrombotic microangiopathy.
- Famciclovir is an oral drug only. Famciclovir is well-tolerated. Compared to acyclovir-TP, penciclovir-TP has lower affinity for viral DNA polymerase but achieves higher intracellular concentrations and has a long intracellular half-life. Famciclovir's pharmacological properties are like acyclovir's.
- Ganciclovir, cidofovir, foscarnet, letermovir, and maribavir are used primarily in the treatment and prophylaxis of CMV retinitis and other CMV end organ infections in immunocompromised patients. Ganciclovir is also effective in the prophylaxis and treatment of HSV but is much more toxic than acyclovir, valacyclovir, and famciclovir.
- Ganciclovir is converted to the monophosphate by a viral kinase called UL97. It is administered intravenously and intravitreally. Oral bioavailability is poor. Valganciclovir is the ganciclovir prodrug for oral administration.
- Ganciclovir's major systemic toxicities include bone marrow toxicity, suppression of fertility, carcinogenicity, and fetotoxicity. Renal toxicity is reported in elderly patients and transplant recipients taking nephrotoxic drugs.

What you need to know and understand

- Cidofovir is a phosphorylated molecule (a nucleotide) activated by host kinases only. The active diphosphate is effective against CMV, HSV, and HPV (human papilloma virus), and adenovirus. It is given IV only along with hydration and probenecid and renally excreted via OATs. Nephrotoxicity is the major dose-limiting toxicity.
- Foscarnet is a phosphonate analog that does not require activation for antiviral activity. It inhibits DNA polymerase, RNA polymerase, and HIV reverse transcriptase. It is given intravenously and renally excreted. Adverse effects are severe, including a high incidence of nephrotoxicity. Other toxicities include hypocalcemia and other electrolyte disturbances, genitourinary ulcerations, headache, hallucinations, and seizures.
- Letermovir's activity is specific to CMV by binding CMV pUL56 and inhibiting the terminase complex required for CMV maturation, packaging, and processing. It is given orally or IV for CMV prophylaxis in allogeneic hematopoietic stem cell transplant recipients.
- Maribavir is a nucleoside analog that is a competitive inhibitor of CMV pUL97 (kinase) for the treatment of post-transplant cytomegalovirus infection that does not respond to the available anti-CMV drugs.

A	Route depends on drug and clinical use Versions for topical / local use are available Poor bioavailability, which decreases with increasing doses ➤ Prodrugs significantly improve systemic levels of the parent drugs.	
	D Wide distribution in body fluids, including CSF	
M	Intracellular conversion to triphosphates	Prodrugs valacyclovir and famciclovir: rapid hepatic conversion to acyclovir and penciclovir, respectively
E	Renal excretion of unchanged drug via glomerular filtration and active secretion • Renal dosing recommendations are listed in drug monographs.	
t_{1/2}	Variable among the different drugs Biological t _{1/2} is related to intracellular accumulation of drug	

Pharmacokinetic Properties: Anti-HSV Drugs

	Acyclovir	Valacyclovir	Famciclovir
A	Oral, IV, topical Bioavailability 10-30% (saturable)	Oral prodrug Acyclovir bioavailability ~55% (peptide transport)	Oral prodrug Penciclovir bioavailability ~75% Achieves higher penciclovir-TP intracellular concentrations than acyclovir-TP
D	Widely distributed including CSF		
E	Glomerular filtration / tubular secretion (OAT) as acyclovir / penciclovir, respectively Renal dosing recommendations are available.		
t _{1/2}	2.5 h adults 4 h neonates ~20 h anuria		2-4 h Prolonged intracellular duration: HSV-1: 10 hrs; HSV-2: 20 hrs VZV: 7 hrs

Anitviral Activity (in vitro) | Efficacy

Acyclovir-TP	HSV-1, HSV-2, VZV VZV: Higher dose required – acyclovir is somewhat less potent against VZV Activity against EBV, CMV, and HHV-6 present but weaker
Famciclovir prodrug: Penciclovir-TP	HSV-1, HSV-2, VZV, EBV, HBV Note: Penciclovir-TP has lower affinity for the viral DNA polymerase than acyclovir-TP, but penciclovir achieves higher intracellular concentrations and has a much longer intracellular half-life.
Selectivity	1) Viral kinase initiates activation of the drug → high concentrations in virus-infected cells but not uninfected cells 2) Drug-TP affinity for viral DNA polymerase higher than for host DNA polymerase.
Efficacy	Acyclovir, valacyclovir, and famciclovir have similar efficacy.

Resistance Mechanisms

Resistance is rare in the immunocompetent host.

Clinically resistant infections reported in up to 10% of immunocompromised host.

Alterations of viral proteins by ***point mutations*** and ***base insertions*** or ***deletions*** in the corresponding genes

1. Mutations in the UL23 gene that encodes viral thymidine kinase → reduced or absent activity
2. Altered thymidine kinase substrate specificity resulting in phosphorylation of thymidine but not acyclovir
3. Altered viral DNA polymerase (rare mechanism of resistance)

Cross-resistance: Thymidine kinase-deficient, acyclovir-resistant HSV:
acyclovir, valacyclovir, famciclovir, ganciclovir

Acyclovir, Valacyclovir and Famciclovir Clinical Considerations

Treatment of HSV and VZV infections

Prophylaxis of HSV, VZV, and CMV infections

Goals of therapy: If initiated during the prodromal stage, anti-HSV/VZV agents may:

- Reduce duration of lesion and acute pain → shorten healing time
- Reduce period of viral shedding → reduces transmission
- VZV: reduce acute neuralgia and post-herpetic neuralgia

Recurrences are common after cessation of therapy and may require long-term suppression.

Therapeutic uses of acyclovir, valacyclovir, famciclovir

Infection	Comments
Genital herpes Systemic therapy Topical antiviral therapy is of marginal benefit and not recommended.	First episodes: • Treatment shortens duration by ~2 days, lesion healing by 4 days, and viral shedding and duration by 7 days. • Treatment does not alter the frequency or severity of recurrent outbreaks. Recurrent genital herpes: • Treatment shortens the overall course by 1-2 days. • Suppressive therapy in patients with frequent recurrence decreases frequency of symptomatic recurrence and asymptomatic viral shedding. Outbreaks may occur when suppression is discontinued.
Herpes labialis (cold sores)	Early treatment reduces duration of lesions by 1-2 days, duration of painful swallowing (odynophagia), and viral shedding are also reduced. • Topical agents require frequent application and are less effective than oral therapy.

Therapeutic uses of acyclovir, valacyclovir, famciclovir

Infection	Comments
Varicella (chicken pox) Herpes zoster (shingles)	Early treatment reduces the total number of lesions, duration of symptoms, viral shedding, and post-herpetic neuralgia: • Varicella: within 24 hours • Herpes zoster: within 72 hours ➤ Practice point: VZV treatment requires higher doses because VZV is less susceptible to the drugs than herpes simplex.
• Neonatal HSV infection • HSV-1 viral encephalitis • Varicella pneumonia • Herpes zoster infection in immunocompromised host	Acyclovir I.V. Early therapy can prevent mortality and complications.
Prophylaxis of CMV infection in allogeneic hematopoietic cell transplant and solid organ transplant recipients	High dose oral acyclovir or valacyclovir Preferred: Ganciclovir, Valganciclovir, Letermovir – see later

Acyclovir, Valacyclovir, Famciclovir Adverse Effects

Safe and Well Tolerated

Side effects (infrequent)
Nausea, vomiting, diarrhea, rash, headache

Acyclovir, valacyclovir, famciclovir

Rare but important toxicities to be aware of in adults and pediatric patients:

Acute kidney injury

- Acyclovir crystal formation → intrarenal obstruction and nephropathy
- Interstitial nephritis due to immune reaction
- Renal tubular necrosis due to direct cytotoxicity by acyclovir metabolite

Acyclovir, intravenous
Slow infusion over 1 hour reduces risk.

➤ **Practice point:** Maintain adequate hydration; the renal effects of acyclovir are uncommon with adequate hydration and avoidance of rapid infusion rates. Monitor BUN, serum creatinine, and urine output.

Neurotoxicity

Confusion, agitation, hallucination, impaired consciousness; rarely, aphasia, ataxia, tremor, myoclonus, seizures

Dose-dependent:
Acyclovir, intravenous
Valacyclovir, high dose

Thrombotic microangiopathy

Thrombotic thrombocytopenic purpura / hemolytic uremic syndrome → organ damage (fatalities have occurred)

Valacyclovir in very high doses for suppressive therapy in HIV/AIDS patients

Pregnancy	Acyclovir for treatment of varicella Acyclovir / Valacyclovir for treatment of genital herpes and for suppressive therapy beginning at 36 weeks gestation in patients with a history of genital lesions
Lactation	Breast feeding is considered acceptable (<10% of relative infant dose appears in breast milk)
Pediatrics	Safety and dosing information are available.
Drug interactions	Mechanisms: Interference with activation by phosphorylation of other drugs Enhanced risk of renal injury with other nephrotoxic drugs Interference with renal excretion

Check your knowledge. You are invited to pause the video and answer the questions.

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What is the pharmacokinetic benefit of the prodrugs of acyclovir and penciclovir?

What adverse effect of intravenous acyclovir may be mitigated by administering by slow infusion and maintaining adequate hydration?

Why are the doses of acyclovir, valacyclovir, and famciclovir higher for treatment of varicella zoster virus than for HSV-1 or HSV-2?

Ganciclovir Pharmacokinetic Properties

A	Ganciclovir I.V.	Valganciclovir (oral) ^{esterases} → Ganciclovir ganciclovir bioavailability ~60%	Intravitreal injection Ophthalmic gel
D	Widely distributed including CSF and intraocular tissues Protein binding 1-2%		
M	Not metabolized (except intracellularly to triphosphate)		
E	Urine, unchanged drug (glomerular filtration and secretion)		
t _{1/2}	~4 hours adults ~3 hours neonates, infants, children Intracellular t _{1/2} ~24 hours Up to 68 hours in patients with severe renal impairment		

Activity (in vitro)	CMV, HSV, VZV, EBV, HHS-6, HHS-8 100x greater activity against CMV compared to acyclovir
Selectivity	1- GCV concentrations are at least 10-fold higher in CMV infected cells than in uninfected cells. 2- GCV is a more potent inhibitor of viral DNA polymerase than of host DNA polymerase (but it is not completely selective). Ganciclovir is more toxic than acyclovir.
Resistance	*1- UL97 gene mutations → ↓ intracellular phosphorylation of ganciclovir (the major mechanism of resistance) 2- Mutations in gene encoding CMV DNA polymerase (UL54 gene)
Cross-resistance	CMV: Ganciclovir dual mutations in UL97 gene → potential cross-resistance with cidofovir; foscarnet variable cross-resistance HSV: Thymidine kinase deficient virus: acyclovir, penciclovir, ganciclovir cross-resistance

Therapeutic Uses Valganciclovir | Ganciclovir

Valganciclovir	CMV retinitis treatment Valganciclovir (oral) is as effective as intravenous ganciclovir. CMV disease prophylaxis Allogeneic stem cell transplant and solid organ transplant recipients Patients with poorly controlled and advanced HIV/AIDS
Ganciclovir I.V.	CMV retinitis treatment and chronic suppression in immunocompromised individuals unable to take or tolerate oral drugs Treatment of CMV tissue invasive disease Prophylaxis of CMV infection in allogeneic stem cell and solid organ transplant recipients
Ganciclovir Intravitreal injection	CMV retinitis in HIV/AIDS patients with immediate sight-threatening disease – in conjunction with systemic therapy VZV retinal necrosis ± intravitreal foscarnet
Ganciclovir ophthalmic gel	Herpetic keratitis treatment

Ganciclovir / Valganciclovir Adverse Effects (not a complete list)

Common side effects	Diarrhea, nausea; pruritis, rash, headache, insomnia
Nephrotoxicity	Acute kidney injury (acute renal failure) potential
➤ Practice points: Ensure adequate hydration / Cautious use in patients receiving nephrotoxic drugs.	
Myelosuppression The major dose-limiting toxicity	Severe leukopenia, neutropenia, anemia, thrombocytopenia, pancytopenia, bone marrow failure with aplastic anemia reported. May occur at any time during treatment. May worsen with continued use. ➤ Practice points: Monitor CBC and platelet counts at baseline and frequently during therapy. Do not use in patients with low absolute neutrophil counts, low platelets, low hemoglobin.
Fertility	Suppression of spermatogenesis, temporary or permanent Suppression of fertility in females
Carcinogenic	Potential to cause cancers in humans
Teratogenic	Potential to cause birth defects in humans

U.S. Boxed Warnings

Check your knowledge. You are invited to pause the video and answer the questions.

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What are ganciclovir's main therapeutic uses?

List ganciclovir's potentially serious adverse effects.

Foscarnet

Pharmacokinetic Properties and Therapeutic Uses

Pharmacokinetics

- I.V. / intravitreal injection
- Distributes to CSF and vitreous humor
Deposition in bone with cumulative IV dosing
- Excreted unchanged in urine via glomerular filtration
- $t_{1/2}$ bimodal
average 3-7 hours; terminal ~88 h
- Sequestration in bone → gradual release over several months

Therapeutic uses

- Immediate sight-threatening CMV retinitis treatment
GCV-foscarnet dual therapy is more effective in delaying progression of retinitis than either drug alone in AIDS patients.
- Acyclovir-resistant HSV and VZV infections
- Preemptive therapy for CMV prophylaxis in allogeneic stem cell transplant recipients
- Treatment of other CMV tissue invasive disease and progressive VZV retinal necrosis (plus ganciclovir)

Foscarnet Major Adverse Effects (not a complete list)

U.S Boxed Warnings

U.S Boxed Warnings	Nephrotoxicity The major toxicity		Renal impairment (Occurs to some degree in the majority of patients)
	➤ Practice points: Frequent monitoring of serum creatinine with dose adjustments for changes in renal function. Ensure adequate hydration / Cautious use in patients receiving nephrotoxic drugs.		
	Seizures	Related to plasma mineral and electrolyte imbalance (Incidence reported in up to 10% of AIDS patients)	
	Mineral and Electrolyte imbalances	Chelation of divalent cations: Hypomagnesemia Hypocalcemia	Also: Hypokalemia Hypo-, hyperphosphatemia
		➤ Practice points: Carefully monitor for changes in plasma mineral and electrolyte changes. Mineral and electrolyte supplementation may be required.	
	Other reported effects	nausea vomiting, anemia, elevated liver enzymes, headache, hallucinations, fatigue	
	Pregnancy, lactation	Safety uncertain	

Cidofovir Pharmacology

- A I.V. only with
Saline prehydration and probenecid
- D CSF levels insignificant
Protein binding <6%
- E Urine, unchanged drug
- $t_{1/2}$ 2-6 hours
Intracellular 17-65 hours
Induction 1x weekly dosing x 2 doses
Maintenance dosing every 2 weeks

- Intracellular phosphorylation by **host** enzymes to cidofovir-diphosphate (2 more phosphate groups are added)
- 1- potent inhibition of viral DNA polymerase by competitive inhibition of deoxycytidine triphosphate (dCTP)
 - 2- incorporation into the viral DNA chain → slows and eventually terminates chain elongation

Therapeutic Uses:

- Treatment: CMV retinitis, tissue invasive disease
- CMV preemptive therapy: hematopoietic cell recipients
- Non-CMV infection treatment: Adenovirus, BK virus, acyclovir-resistant HSV, nonvariola orthopoxvirus – mpox (monkeypox) with tecovirimat

Cidofovir Major Adverse Effect (not a complete list)

U.S. Boxed Warnings	Nephrotoxicity	The major toxicity – See next slide
	Neutropenia	Monitor neutrophil counts during therapy
	Carcinogenic	Potential to cause cancers in humans
	Teratogenic	Potential to cause birth defects in humans
	Fertility	Suppression of spermatogenesis, temporary or permanent
	Ocular complications	Uveitis, iritis Iritis, intraocular hypotonia, permanent vision impairment potential
	Contraindications	Other nephrotoxic agents concurrently or within 7 days of cidofovir administration Direct intraocular administration
	Pregnancy	Not recommended
	Drug interactions	Nephrotoxic agents Probenecid inhibits renal OAT

Cidofovir Nephrotoxicity Manifestations and Management

Nephrotoxicity:

Proximal tubular dysfunction

The chief dose-limiting side effect

Cases of acute renal failure resulting in dialysis and/or contributing to death have occurred with as few as 1 or 2 doses of cidofovir

Management essential:

Probenecid and intravenous saline hydration prior to infusion of cidofovir

Manifestations:

Proteinuria, azotemia (elevated blood urea nitrogen), glycosuria, metabolic acidosis, Fanconi syndrome

Rationale for saline and probenecid:

Saline hydration increases the volume of fluid flowing through the lumen of the nephron and inhibition of organic anion transporters (OAT) slows entry of cidofovir into the proximal tubule.

Check your knowledge. You are invited to pause the video and answer the questions. This slide is not in the video.

Foscarnet is a structural analog of what?

Thinking about its structure, what is foscarnet's mechanism of action?

Thinking about its PK profile and mechanism of action, what are foscarnet's adverse effects?

What is foscarnet's spectrum of activity?

Check your knowledge. You are invited to pause the video and answer the questions. This slide is not in the video.

Cidofovir is a structural analog of what?

Thinking about its structure, what is cidofovir's mechanism of action?

What is cidofovir's major dose-limiting adverse effect and how is it mitigated?

What is cidofovir's spectrum of activity?

Letermovir Pharmacokinetics

All parameters are in hematopoietic stem cell transplant patients.

A Oral (preferred route)
I.V.
(hydroxypropyl betadex vehicle)

Practice point:

Monitor CrCl: Vehicle in IV solution may accumulate in patients with impaired renal function.

D 99% plasma protein binding
Substrate of transporters

P-glycoprotein
OATP1B1/1B3 (hepatic transporters)

Bioavailability
35% but increases to 85% when
given with cyclosporine likely due to
cyclosporine inhibition of OATP

Practice point:

Consult dosing recommendations when using
letermovir with cyclosporine.

Cyclosporine is a maintenance immunosuppressant frequently administered to transplant recipients to prevent rejection.

M Minor hepatic metabolism via
glucuronidation

Inhibits CYP3A4, P-gp, OATP
Induces CYP2C9, 2C19

E Feces, 70% unchanged drug

$t_{1/2}$ 12 h

Letermovir Therapeutic Use and Adverse Effects

- Prophylaxis of CMV infection and disease in adult CMV-seropositive recipients of an allogeneic hematopoietic stem cell transplant
- Activity: Letermovir is highly specific for and highly active against CMV, including drug-resistant isolates.
- It has no activity against other viruses.

- Nausea, vomiting, diarrhea, abdominal pain
- Headache, fatigue
- Peripheral edema
- Tachycardia
- Decreased platelet count
- Decreased hemoglobin

Maribavir Pharmacology

PK	Oral, protein binding 98%, CYP3A4 metabolism, excreted in urine/feces as inactive metabolites
MOA	Competitive inhibitor of pUL97 (CMV kinase) → prevents phosphorylation of CMV proteins → inhibits CMV DNA replication, encapsidation, and nuclear egress
Resistance	Substitutions in UL97 gene confer resistance to maribavir, ganciclovir, valganciclovir
Use	Treatment of post-transplant CMV infection not responsive to the available anti-CMV drugs in adults and children ≥ 12 years old and ≥ 35 kg
AEs	Diarrhea, nausea, vomiting, decreased hemoglobin and platelet count, fatigue, taste disorder, increased serum creatinine
Pregnancy	Adverse effects observed in animal reproduction studies at doses lower than equivalent recommended doses; not tested in pregnant humans
DDIs	Avoid strong CYP3A4 inducers and inhibitors Do not combine with ganciclovir due to diminished efficacy of ganciclovir

Check your knowledge. You are invited to pause the video and answer the questions. This slide is not in the video.

What is pUL56?

How does blocking pUL56 prevent viral replication?

What is the name of this anti-pUL56 drug and its spectrum of activity?

What is the therapeutic indication of this drug?

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What are the mechanisms of drug-drug interactions with this anti-UL56 agent?

Check your knowledge. You are invited to pause the video and answer the questions. This slide is not in the video.

What is pUL97?

How does blocking pUL97 prevent viral replication?

What is the name of this anti-pUL97 drug and its spectrum of activity?

What is the therapeutic indication of this drug?

What are the mechanisms of drug-drug interactions with this anti-UL97 agent?

Drug → active form	Structure	Mechanism	Antiviral Activity
Acyclovir → acyclovir TP (HSV thymidine kinase → ACV-MP)	acyclic guanosine analog (lacks 3' hydroxyl group)	Competitive inhibition of dGTP DNA pol inhibition, <i>obligate</i> chain termination	HSV-1, HSV-2, VZV CMV – weak activity
Valacyclovir →...→ acyclovir TP	acyclovir prodrug	as above	as above
Famciclovir →...→ penciclovir TP (HSV thymidine kinase → PCV-MP)	penciclovir prodrug acyclic guanosine analog	Competitive inhibition of dGTP DNA pol inhibition, eventual chain termination	HSV-1, HSV-2, VZV
Ganciclovir → ganciclovir TP (CMV UL97 kinase → GCV-MP)	acyclic guanosine analog	Competitive inhibition of dGTP DNA pol inhibition, eventual chain termination	CMV, HSV, VZV other herpes viruses
Valganciclovir →...→ ganciclovir TP	ganciclovir prodrug	as above	as above
Foscarnet (active)	pyrophosphate analog	Noncompetitive block of pyrophosphate(PPi) binding site on viral DNA pol → inhibits cleavage of PPi from deoxyTP nucleotides	CMV, HSV, VZV, other herpes viruses, HIV reverse transcriptase
Cidofovir → cidofovir diphosphate (host kinases, not viral kinase)	acyclic cytidine analog (phosphate on structure)	Competitive inhibition of dCTP DNA pol inhibition, eventual chain termination	Broad spectrum: Herpesviruses and others
Letermovir (active)	synthetic small molecule	Binds CMV pUL56 → inhibits terminase complex → inhibits maturation, packaging, and termination process in CMV	CMV only
Maribavir (active)	benzimidazole riboside	Competitive inhibitor of CMV pUL97 → prevents phosphorylation of CMV proteins → inhibits CMV DNA replication, encapsidation, and nuclear egress	CMV only

Drug	PK	Therapeutic Use	Adverse effects	Comments
Acyclovir	Oral, IV, topical; widely distributed, urine/unchanged drug Bioavailability 10-30% t _{1/2} ~2.5 h	HSV-1, HSV-2, VZV treatment, prophylaxis CMV prophylaxis in low-risk transplant recipients	Well tolerated (nfrequent GI, headache) Rare but serious with high dose: crystalluria and neurotoxicity	Slow IV infusion Renal dosing required
Valacyclovir	Oral, rapid hepatic metabolism to acyclovir Bioavailability 55-70%	HSV-1, HSV-2, VZV treatment and prophylaxis CMV prophylaxis in low-risk transplant recipients	Well tolerated (infrequent GI headache); Thrombotic microangiopathy reported (rare)	Renal dosing
Famciclovir	Oral, rapid hepatic metabolism to penciclovir Bioavailability 75% Urine/unchanged drug	HSV-1, HSV-2, VZV treatment, prophylaxis	Well tolerated (nfrequent GI, headache)	Prolonged intracellular half-life of active drug-TP; Renal dosing
Ganciclovir	IV, intravitreal, topical; widely distributed including intraocular tissues; urine/unchanged drug t _{1/2} ~4 h	Treatment: CMV sight-threatening retinitis and other tissue invasive disease Prophylaxis: CMV in allogeneic stem cell and solid organ transplant recipients Treatment: VZV retinal necrosis Topical: herpetic keratitis	Significant toxicity Myelosuppression (major dose-limiting toxicity), carcinogenic, suppression of fertility, embryotoxic, nephrotoxicity when given with other nephrotoxic drugs	Renal dosing
Valganciclovir	Oral Hepatic / mucosal conversion to ganciclovir	Treatment: CMV retinitis, esophagitis or colits Prophylaxis: CMV in stem cell and solid organ transplant recipients	As for ganciclovir	As effective as IV ganciclovir

Drug	PK	Therapeutic Use	Adverse effects	Comments
Foscarnet	IV, intravitreal; widely distributed; urine/unchanged drug t _{1/2} 3-7 h; terminal 88h ave	Treatment: sight-threatening CMV retinitis plus GCV CMV tissue invasive disease Acyclovir-resistant HSV, VZV VZV retinal necrosis plus GCV Prophylaxis: allogeneic stem cell transplant recipients	Nephrotoxicity (major toxicity) Seizures Mineral, electrolyte imbalance Nausea, vomiting, anemia, elevated LFTs, headache, hallucinations, QT interval prolongation	May sequester in bone Pregnancy, lactation safety uncertain
Cidofovir	IV only, low CSF levels; urine/unchanged drug t _{1/2} 2-6 h prolonged intracellular t _{1/2}	Treatment: CMV retinitis and tissue-invasive disease Preemptive tx allogeneic stem cell transplant recipients Non-CMV infections: adenovirus, BK virus, resistant HSV, mpox (with tecovirimat)	Major: Nephrotoxicity, proximal tubular dysfunction Neutropenia, carcinogenicity, fertility suppression, teratogenicity, ocular complications	Administer with prehydration and probenecid to reduce nephrotoxicity
Letermovir	Oral, IV (in vehicle to enhance solubility) Protein binding 99% Minimal hepatic metabolism Feces/unchanged drug	Preemptive tx allogeneic stem cell transplant recipients	GI, headache, fatigue, tachycardia, peripheral edema, decreased platelet and hemoglobin count	High potential for DDIs: Substrate of OATPs, P-gp Inhibits CYP3A4, OATPs, P-gp Dose adjustment with cyclosporine
Maribavir	Oral only Protein binding 98% CYP3A4 metabolism t _{1/2} ~4 h (transplant pts)	Post-transplant CMV infection refractory to all other agents	GI, decreased platelet and hemoglobin count, increased SeCr	Avoid combination with strong CYP3A4 inducers and ganciclovir (may ↓GCV effect)

Summary

- Most anti-herpes drugs are nucleoside analogs activated by viral and host kinases; the triphosphates inhibit viral DNA polymerase.
- Acyclovir, valacyclovir, and famciclovir, which are used in the treatment and prophylaxis of HSV and VZV, are well tolerated. Acyclovir intravenous is associated with CNS adverse effects and may cause nephrotoxicity – crystalline nephropathy, interstitial nephritis.
- Ganciclovir, valganciclovir, foscarnet, cidofovir, letermovir, and maribavir are used in the treatment and prophylaxis of CMV retinitis and other CMV end organ disease. These drugs are associated with significant toxicities.
- Letermovir inhibits the CMV DNA terminase subunit pUL56 required for DNA packaging and processing and thus replication; this component of the terminase subunit has no equivalent target enzyme in the human body. It is indicated for CMV prophylaxis in adult CMV-seropositive recipients of an allogeneic hematopoietic stem cell transplant. It is available for oral and IV administration, CYP3A4 metabolism. Letermovir-resistant isolates with UL56 mutation have been reported during therapy. Side effects include

Summary of viral resistance mechanisms.

Acyclovir Valacyclovir Famciclovir	UL23: HSV thymidine kinase mutations that result in 1) functionally deficient enzyme or 2) kinase that decreases phosphorylate the drug HSV-1 and HSV-2 cross-resistance to acyclovir, valacyclovir, famciclovir, ganciclovir	HSV DNA polymerase mutations that reduce affinity for the drug, which reduces the efficacy of the drug
Ganciclovir Valganciclovir	UL97: CMV kinase mutations that decreases phosphorylation of the drug (major mechanism of resistance)	UL54: CMV DNA polymerase mutations that alter binding site, which reduces the drug's efficacy.
	UL97 mutations increase the likelihood of CMV cross-resistance to ganciclovir, cidofovir, maribavir.	Potential for CMV cross-resistance to foscarnet related to UL54 mutations is variable.
Foscarnet	Foscarnet does not require UL97 activity because it is not require phosphorylation for activity.	UL54 mutations – as above
Cidofovir	Cidofovir does not require UL97 as above	UL54 mutations – as above
Letermovir	UL56 component of the terminase complex: Mutations in the UL56 protein (pUL56) reduce affinity for the drug, resulting in reduced efficacy. Isolates have been reported during therapy.	
Maribavir	UL97: CMV kinase mutations reduce affinity for maribavir, resulting in reduced efficacy. Cross-resistance to ganciclovir.	

Check your knowledge. You are invited to pause the video and answer the questions.

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What is the pharmacokinetic benefit of the prodrugs of acyclovir and penciclovir?

- Valacyclovir and famciclovir provide much better bioavailability of acyclovir and penciclovir.

What adverse effect of intravenous acyclovir may be mitigated by administering by slow infusion and maintaining adequate hydration?

- The risk of interstitial nephritis and renal tubular necrosis due to acyclovir crystal formation may be reduced

Why are the doses of acyclovir, valacyclovir, and famciclovir higher for treatment of varicella zoster virus than for HSV-1 or HSV-2?

- VZV is less susceptible to the drugs, therefore higher doses are needed.
- Said another way, HSV is more susceptible, therefore lower doses can be used.

Check your knowledge. You are invited to pause the video and answer the questions.

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What are ganciclovir's main therapeutic uses?

- CMV retinitis: Valganciclovir orally. Intravitreal ganciclovir with oral valacyclovir for immediate sight-threatening retinitis in AIDS patients.
- Treatment and chronic suppression of CMV infection in immunosuppressed patients and CMV prevention in those receiving hematopoietic cell or solid organ transplant.

List ganciclovir's potentially serious adverse effects.

- Bone marrow suppression – dose-limiting toxicity, spermatogenesis suppression and suppression of fertility in females, fetal toxicity (based on animal data), carcinogenicity (based on animal studies)
- Potential for acute renal injury – Ensure adequate hydration.

Check your knowledge. You are invited to pause the video and answer the questions.

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Foscarnet is a structural analog of what?

- An inorganic pyrophosphate analog

Thinking about its structure, what is foscarnet's mechanism of action?

- Foscarnet mimics the pyrophosphate product of DNA polymerization
- Meaning: In DNA synthesis, a pyrophosphate (PPi) is released as a byproduct when a new nucleotide is added to the DNA strand by DNA polymerase. Hydrolysis of the PPi (Pi + Pi) creates energy that helps drive the DNA synthesis reaction forward – it is an energetically favorable process.

Thinking about its PK profile and mechanism of action, what are foscarnet's adverse effects?

- Direct renal tubular toxicity with necrosis (excreted unchanged by glomerular filtration), precipitation in renal tubules (crystal formation), electrolyte imbalances due to chelation of divalent cations such as Ca^{2+} and Mg^{2+} that can lead to further renal impairment, and dehydration that can exacerbate renal toxicity. And other adverse effects.

What is foscarnet's spectrum of activity?

- Foscarnet has a broad spectrum of activity (in vitro) against multiple viruses in the Herpesviridae family.

Check your knowledge. You are invited to pause the video and answer the questions. This slide is not in the video.

Cidofovir is a structural analog of what?

- An acyclic cytidine monophosphate analog

Thinking about its structure, what is cidofovir's mechanism of action?

- Notice: Cidofovir is already phosphorylated so does not need to be phosphorylated to a monophosphate by viral kinase. Host kinases phosphorylate it two more times to active cidofovir “diphosphate” ($1 + 2 = 3$ phosphates).
- The active cidofovir diphosphate inhibits viral DNA polymerase by competing with dCTP and by incorporation into the viral DNA chain. DNA chain elongation slows and eventually stops.

What is cidofovir's major dose-limiting adverse effect and how is it mitigated?

- Renal tubular toxicity. Cidofovir is excreted unchanged and taken up by renal tubular cells via organic anion transporters, leading to acute tubular necrosis. Precipitation in renal tubules (crystal formation) leads to obstruction and induction of apoptosis contributes to renal dysfunction.
- Administration of probenecid (blocks OATs) and saline hydration (increases tubular fluid) are given to reduce the risk of nephrotoxicity.

What is cidofovir's spectrum of activity?

- Cidofovir is a broad spectrum antiviral drug.

Check your knowledge. You are invited to pause the video and answer the questions. This slide is not in the video.

What is pUL56?

- A protein in the CMV terminase complex (pUL51, pUL56, pUL89) and the target of letermovir.

How does blocking pUL56 prevent viral replication?

- Binding to pUL56 inhibits the activity of the terminase complex – maturation, packaging, and the termination process are inhibited.

What is the name of this anti-pUL56 drug and its spectrum of activity?

- Letermovir is highly specific for CMV. It is not active against other viruses, including others in the Herpesviridae family.

What is the therapeutic indication of this drug?

- Prophylaxis of CMV infection and disease in adult CMV-seropositive recipients of an allogeneic hematopoietic stem cell transplant.

What are the mechanisms of drug-drug interactions with this anti-UL56 agent?

- **High potential for adverse drug events!**
- Letermovir inhibits CYP3A4, P-gp, OATP1B1, OATP1B3 and induces CYP2C9 and 2C19.

Check your knowledge. You are invited to pause the video and answer the questions. This slide is not in the video.

What is pUL97?

- A protein kinase that phosphorylates CMV proteins (and phosphorylates ganciclovir to ganciclovir monophosphate).

How does blocking pUL97 prevent viral replication?

- Blocking the activity of pUL97 prevents phosphorylation of CMV proteins, which inhibits CMV DNA replication, encapsidation, and exit from the host nucleus.

What is the name of this anti-pUL97 drug and its spectrum of activity?

- Maribavir is specific for CMV, including strains refractory to ganciclovir, foscarnet, and cidofovir. It does not exhibit significant activity against other viruses.

What is the therapeutic indication of this drug?

- Treatment of post-transplant CMV infection refractory to treatment with ganciclovir, valganciclovir, foscarnet, and cidofovir in adults and children ≥ 12 years old and ≥ 35 kg.

What are the mechanisms of drug-drug interactions with this anti-UL97 agent?

- Maribavir is a CYP3A4 substrate. Strong CYP3A4 inducers can decrease maribavir plasma concentrations and should be avoided. Maribavir may increase plasma concentrations of other CYP3A4 substrates.